



LAB 1

GEOSPATIAL REPRESENTATION LEARNING

Lab 1: Why Geospatial Representation Learning?

PRESS – OR SPACE. →

Lab 1 Overview

Learning Objectives

By the end of this lab, you will be able to:

1. Understand the mathematical intuition behind PCA and t-SNE
2. Explain your assigned method to a peer who studied the other one
3. Compare the two approaches and discuss when each is appropriate

Expert Group Activity

This lab uses a **jigsaw learning** structure.

Group A — PCA

Study **Principal Component Analysis** as a way to reduce high-dimensional geospatial data to a 2D embedding.

→ You will teach Group B what PCA does and when to use it.

Group B — t-SNE

Study **t-distributed Stochastic Neighbor Embedding** as a way to visualize geospatial feature spaces.

→ You will teach Group A what t-SNE does and when to use it.

Peer Teaching Session

After studying your method, form pairs with one person from Group A and one from Group B.

Each person explains (~5 min):

1. What your method does — in plain language
2. The key mathematical idea behind it
3. One strength and one limitation for geospatial use

Together, discuss (~5 min):

- Which method produced cleaner clusters in the examples you studied?
- Could you combine them? (e.g. PCA → t-SNE)
- Which would you use for a downstream ML task — and why?

Submit Your Exam Questions

Based on what you learned and discussed, write **2 exam questions** — one on PCA and one on t-SNE.

PCA question

Write a question that tests conceptual understanding — not just a definition.

Example style: "Under what conditions would PCA fail to separate two land-cover classes that t-SNE separates cleanly?"

t-SNE question

Write a question that probes a common misconception or a key limitation.

Example style: "A student trains t-SNE on a dataset and wants to embed new satellite patches. What is the problem with this approach?"

Submit via the course portal — best questions will appear in the quiz!

Live Quiz

We'll work through the best submitted questions together as a class.

Format: Question appears on screen → 60 seconds to think → discuss with your neighbour → class vote / discussion

Goal: Surface misconceptions and reinforce the key distinctions between PCA and t-SNE before the next lecture.